

✓ **Problem 1.** Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(x^2) - f(y^2) \leq (f(x) + y)(x - f(y))$$

holds for all $x, y \in \mathbb{R}$.

✓ **Problem 2.** Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(xy) \leq yf(x) + f(y)$$

holds for all $x, y \in \mathbb{R}$.

✓ **Problem 3.** Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$xf(y) + f(x + y) \geq (y + 1)f(x) + f(y)$$

holds for all $x, y \in \mathbb{R}$.

Problem 4. Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$(x - y)(f(x) + f(y)) \leq f(x^2 - y^2)$$

holds for all $x, y \in \mathbb{R}$.

Problem 5. Find all functions $f : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}$ such that

$$x + 2 \max\{y, f(x), f(z)\} \geq f(f(x)) + 2 \max\{z, f(y)\}$$

holds for all $x, y, z \in \mathbb{R}_{\geq 0}$.

Problem 6. Prove that there is no such function $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(x - f(y)) \leq x - yf(x)$$

holds for all $x, y \in \mathbb{R}$.

Problem 7. Let $\mathbb{R}^+$ denote the set of positive real numbers. Find all functions $f : \mathbb{R}^+ \rightarrow \mathbb{R}^+$ such that for each $x \in \mathbb{R}^+$, there is exactly one $y \in \mathbb{R}^+$ satisfying

$$xf(y) + yf(x) \leq 2$$

Problem 8. A function $f : \mathbb{R} \rightarrow \mathbb{R}$ satisfies

$$f(x + y) \leq yf(x) + f(f(x))$$

for all $x, y \in \mathbb{R}$. Prove that $f(x) = 0$ for all $x \leq 0$.

Problem 9. Let $\mathbb{R}_{>0}$ be the set of positive real numbers. Determine all functions $f : \mathbb{R}_{>0} \rightarrow \mathbb{R}_{>0}$ such that

$$x(f(x) + f(y)) \geq (f(f(x)) + y)f(y)$$

for every $x, y \in \mathbb{R}_{>0}$.



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